The Impact of Digital Rights Management (DRM) on Consumer Rights: A Call for Reform

Abstract

This policy position paper analyses the effects of Digital Rights Management (DRM) on consumer rights in the digital marketplace. It contends that present DRM implementations have established a considerable imbalance, disproportionately favouring copyright holders to the detriment of genuine consumer interests. This paper analyses landmark legal cases such as Vernor v. Autodesk and Capitol Records v. ReDigi, industry practices like Tor Books' DRM-free model, and the challenges encountered by libraries and educational institutions, illustrating how DRM frequently surpasses the limits of copyright protection to curtail consumer autonomy and access. Based on empirical evidence challenging the efficacy of DRM in combating piracy, the paper advocates for policy reforms to restore balance: instituting non-negotiable minimum consumer rights, augmenting transparency mandates, formulating a digital first sale doctrine, promoting less restrictive models, bolstering protections for libraries, and financing independent research. These ideas would establish a more equitable ecosystem advantageous to artists, customers, and society at large.

Introduction

In the modern digital era, content consumption has progressively transitioned from conventional physical media to digital ones. This shift has led to substantial advancements in the techniques of safeguarding intellectual property. Digital Rights Management (DRM) technologies developed as a mechanism for copyright holders to regulate the use, alteration, and dissemination of digital content. This transformation has unintentionally generated a complex disparity between safeguarding copyright holders' interests and upholding consumer rights.[1] This policy position paper analyses the effects of DRM on consumer rights, highlighting how existing implementations disproportionately favour copyright holders, and suggests measures to re-establish balance in the digital marketplace.

The target audience of this paper includes lawmakers as well as consumer groups together with officials who determine copyright laws and DRM protocol definitions. The continuing rapid growth of digital content consumption makes it a critical priority to solve the current problems with DRM implementation procedures. Modern policies establish tight restrictions on how consumers can use digital content which results in customers possessing less ownership rights than physical formats.[2] E-books controlled by Digital Rights Management (DRM) prevent all potential actions included with traditional books such as lending them and gift-giving their ownership.

This paper investigates multiple critical matters regarding DRM and customer ownership rights. First, the paper examines how DRM fundamentally changes digital ownership by replacing actual ownership with restrictive licensing models in the digital space. Secondly it evaluates important judicial decisions which showcase the copyright protection-clash-with-consumer rights conflicts specifically through Capitol Records, LLC v. ReDigi Inc. and Vernor v. Autodesk, Inc. The analysis also investigates industrial standards alongside non-restrictive DRM solutions with Tor Books as an illustration. The paper also evaluates how Digital Rights Management systems affect libraries and educational establishments that provide crucial access to information along with cultural materials.

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[1] Mir Mohammad Azad, Abu Hasnat Shohel Ahmed, & Asadul Alam, Digital Rights Management, 10 INT'L J. COMPUT. SCI. & NETWORK SOC'Y 24, 24-33 (2010).
[2] L. Jean Camp, DRM: Doesn't Really Mean Digital Copyright Management, KSG Working Paper Series RWP02-034 (Aug. 2002), https://doi.org/10.2139/ssrn.348941.

The relationship between these factors displays troubling trends because DRM systems frequently go beyond their authorized copyright domains thereby diminishing consumer freedoms and invading their privacy while obstructing traditional copyright-law approved activities. These issues matter greatly to our audience because they impact not only individual consumer experiences but also broader societal interests in knowledge dissemination, innovation, and cultural participation. The development of a prosperous digital content system requires pursuing DRM strategies which protect copyright holder needs alongside consumer freedoms.

Key Issues and Positions

DRM and Digital Ownership

Digital Rights Management significantly transforms the concept of ownership in the digital sphere. The implementation of DRM on digital content through restrictive licenses changes how consumers acknowledge ownership because ownership rights shift when purchasing digital content.[3] When DRM takes over it establishes a major change from standard property rights which leads users to surrender essential rights they commonly had with physical products.

The case of Vernor v. Autodesk, Inc. (2010) provides a perfect example of this transformation. The court ruled that consumers who purchase Autodesk software do not actually own it but instead merely acquire a limited license to use it in specific ways as defined within the End User License Agreement (EULA).[4] Consequently, they cannot resell the software, despite having paid for it. The court's decision reinforced the power of EULAs to define the terms and conditions under which software can be used, effectively prioritizing contractual terms over traditional ownership rights.

This case demonstrates how DRM transforms ownership from a broad right to a limited privilege. For many consumers, this limitation feels fundamentally unfair. People typically expect to own what they purchase without excessive constraints on how they use it. This disconnect between consumer expectations and legal reality is particularly acute for digital goods such as e-books and music. Most people believe they should be able to give away or trade their digital purchases in the same manner as physical products, but DRM technology, backed by legal

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[3] Karen Coyle, The Technology or Rights: Digital Rights Management, Part 4, under "DRM and the Law" (based on a talk originally given at the Library of Congress, Nov. 19, 2003), https://www.kcoyle.net/drm_basics4.html.
[4] Vernor v. Autodesk, Inc., 621 F.3d 1102 (9th Cir. 2010).

precedents like Vernor, prevents these uses. Consequently, consumers end up with fewer rights than they anticipate when making digital purchases.[5]

From a policy perspective, this transformation raises significant concerns about market transparency and consumer understanding. Many consumers do not fully comprehend that when they "buy" digital content, they are actually purchasing a limited license rather than obtaining ownership rights comparable to physical goods. This lack of transparency can be seen as a form of market failure. This is because consumers cannot make fully informed purchasing decisions without being able to clearly understand the limitations that accompany their digital purchases.[6]

Furthermore, the shift from ownership to licensing has a lot of significant implications for cultural preservation as well as access. Unlike physical books that can be preserved in libraries for future generations, DRM-protected digital content may easily become inaccessible if the provider discontinues their support or goes out of business. This creates a rather precarious situation for digital cultural heritage and at the same time, it contradicts the longstanding social values regarding the preservation of knowledge as well as creative works.[7]

Legal Cases and Consumer Rights

Many landmark legal cases have shaped the DRM and consumer rights landscape, revealing a consistent pattern of prioritizing copyright protection over the interests of consumers. These cases highlight the tendency of the legal system to favor the holders of the corporate rights at the expense of individual consumers as well as illustrate the need for policy reform.

In Capitol Records, LLC v. ReDigi Inc., ReDigi attempted to create a marketplace for users to resell their digital music files, ensuring that when a file was sold, the original copy was erased from the seller's computer.[8] This approach sought to replicate for digital goods the first sale doctrine that allows the resale of physical goods. However, Capitol Records argued that transferring digital files necessarily involved making another copy, violating copyright law. The

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[5] Frederick W. Dingledy & Alex Berrio Matamoros, What Is Digital Rights Management?, in DIGITAL RIGHTS MANAGEMENT: THE LIBRARIAN'S GUIDE 1, 1-25 (Catherine A. Lemmer & Carla P. Wale eds., 2016).
[6] Amber Sami Kubesch & Stephen Wicker, Digital Rights Management: The Cost to Consumers, 103 PROC. OF THE IEEE 726, 726-33 (May 2015), https://doi.org/10.1109/JPROC.2015.2418457.
[7] Truls Fretland, Lothar Fritsch, & Arne-Kristian Groven, State of the Art in Digital Rights Management, MARIAGE Project Report 1018 (Norsk Regnesentral [Norwegian Computing Center], 2008).
[8] Capitol Records, LLC v. ReDigi Inc., 934 F. Supp. 2d 640 (S.D.N.Y. 2013).

court agreed with Capitol Records, holding that the first sale doctrine does not apply to digital products. This decision significantly impacted consumers by establishing that digital products are governed differently than physical ones, with consumers having fewer rights than they might have expected.

The case of MDY Industries, LLC v. Blizzard Entertainment, Inc. involved a tool called "Glider" that helped players in the online game World of Warcraft perform certain actions more efficiently.[9] Blizzard claimed that this tool violated their DRM and was illegal under the Digital Millennium Copyright Act (DMCA). The court sided with Blizzard, ruling that Glider circumvented the game's copy protection even though it did not involve direct piracy. This case is significant because it demonstrates how companies use DRM not only to prevent unauthorized copying but also to control how people use their products after purchase. For consumers, this raises important questions: If you paid for a game, shouldn't you be able to use it however you want? Critics of DRM have expressed concerns that such cases represent excessive corporate control over products legally purchased by consumers.

In HarperCollins Publishers LLC v. Open Road Integrated Media, Inc., the dispute centered on e-book publishing rights.[10] HarperCollins sued Open Road for offering an e-book of the novel "Julie of the Wolves," arguing that only HarperCollins had the right to distribute the book in any form, including digital, despite the fact that the original agreement did not explicitly include digital rights (as e-books did not exist at the time of contract signing). The court agreed with HarperCollins, holding that digital rights were included in the agreement. This case illustrates how ambiguous contractual provisions can become problematic when new technologies emerge. For consumers, it raises issues regarding e-book availability and access, as many e-books incorporate DRM that restricts sharing or lending, making them less transferable than physical copies.

A common thread across these cases is that DRM and copyright laws generally favor producers and restrict consumers' legal rights. The Capitol Records v. ReDigi decision established that digital copies cannot be traded like physical copies. The Blizzard case showed how DRM can be used not just to prevent copying but also to regulate how people use products in ways that don't

[9] MDY Industries, LLC v. Blizzard Entertainment, Inc., 629 F.3d 928 (9th Cir. 2010).
[10] HarperCollins Publishers LLC v. Open Road Integrated Media, Inc., 754 F.3d 132 (2d Cir. 2014).

necessarily harm the company. The HarperCollins case demonstrated how DRM in e-books makes it harder to obtain and share digital content. Many consumers find these restrictions unfair, believing they should have greater freedom with products they've purchased. But in the digital world, laws and DRM often treat digital purchases more like leases, where companies retain control over how content can be used.[11]

These cases underscore a significant imbalance in the current legal framework governing digital content and DRM. While copyright law aims to strike a balance between creators' and users' interests, the current application of DRM, bolstered by judicial decisions, has disrupted this equilibrium. As digital content continues to grow in importance, these issues will only become more prominent, necessitating policy interventions to restore a more balanced approach.

Industry Practices and Alternatives

Despite the prevalence of restrictive DRM in the digital marketplace, some industry players have explored alternative approaches that better balance copyright protection with consumer rights. These examples offer valuable insights into potential policy reforms and demonstrate that less restrictive approaches can be economically viable.

A notable example is Tor Books, a division of Macmillan, which made headlines in 2012 when it removed DRM from all its e-books. This decision came after considering feedback from readers and authors who were frustrated with the DRM restrictions they faced.[12] Tor's experience is significant as it demonstrates the feasibility of DRM-free approaches. After removing DRM, the publisher reported no increase in piracy, contradicting one of the primary fears in the industry. Instead, it created a better relationship between consumers and products, as customers appreciated being able to use their e-books on any device they wanted without restrictions.

Tor's decision was prompted by author and reader criticism. Many readers of science fiction and fantasy—genres with highly engaged audiences—had expressed their displeasure with DRM. They argued that once they purchased an e-book, they should own it outright, similar to a printed book. Authors, including prominent figures like Cory Doctorow, had long advocated for DRM-

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[11] Jonathan Zittrain, The Copyright Cage, LEGAL AFFAIRS, July/Aug. 2003.
[12] Julie Crisp, One Year Later, the Results of Tor Books UK Going DRM-Free, TOR (Apr. 29, 2013), https://www.tor.com/2013/04/29/tor-books-uk-drm-free-one-year-later.

free publishing. Doctorow has consistently argued that DRM doesn't effectively prevent piracy but instead punishes paying consumers while pirates find ways around it regardless.[13]

Another motivation behind Tor's decision was that DRM often led to platform locking. For instance, a reader who purchased an e-book from Amazon's Kindle store couldn't easily transfer it to an Apple iPad or a Nook e-reader due to compatibility issues. This effectively locked readers into a single ecosystem, which many found restrictive. By eliminating DRM, Tor ensured that readers could buy a book once and read it on any device they preferred.[14]

Despite the fears that removing DRM would unleash uncontrolled piracy, Tor Books experienced no noticeable increase in piracy after the implementation of their DRM-free policy. Sales remained steady, and in some cases, customer satisfaction as well as customer loyalty increased. The freedom customers received to control their legally bought books created satisfaction which led them to maintain their commitment toward Tor's authors. This outcome confirmed what many digital rights advocates had argued: piracy is not necessarily due to the absence of DRM but often results from inconvenient access to legal content. Through its improved system of making books available to paying readers Tor proved to be more effective at lowering piracy motives than creating more opportunities for piracy.[15]

Despite Tor's success, most major publishers have been reluctant to embrace DRM-free policies. Publishing giants such as HarperCollins, Penguin Random House, and Hachette continue to enforce DRM, arguing that widespread file-sharing would damage their revenue streams. For these companies, DRM isn't just about preventing piracy; it's also about maintaining control over distribution. With DRM, publishers can restrict how and where their books are accessed, ensuring readers remain within certain platforms like Amazon Kindle or Apple Books. This control facilitates monitoring sales, enforcing licensing agreements, and maintaining consistent pricing.[16]

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[13] Cory Doctorow, Audio Version of Cory Doctorow's DRM Talk (talk to Microsoft Research Group at its Redmond offices on June 17, 2004), mp3 audio, 39:47, https://kottke.org/04/06/cory-drm-talk.

[14] Priti Trivedi, Writing the Wrong: What the E-book Industry Can Learn from Digital Music's Mistakes with DRM, 18 J.L. & POL'Y 925, 935 (2010), https://brooklynworks.brooklaw.edu/jlp/vol18/iss2/10.

[15] Tim O'Reilly, Piracy Is Progressive Taxation, and Other Thoughts on the Evolution of Online Distribution, O'REILLY (2002), https://www.oreilly.com/ideas/piracy-is-progressive-taxation-and-other-thoughts-on-the-evolution-of-online-distribution.

[16] Stefan Bechtold, From Copyright to Information Law: Implications of Digital Rights Management, in SECURITY AND PRIVACY IN DIGITAL RIGHTS MANAGEMENT, LECTURE NOTES IN COMPUTER SCIENCE, vol. 2320, 213 (Tomas

However, some smaller publishers have followed Tor's lead. Companies like Baen Books and web stores like Humble Bundle have experimented with DRM-free content, demonstrating that it's possible to profitably sell e-books without restrictions. Similar shifts have occurred in other media sectors. GOG.com (Good Old Games) adopted a DRM-free stance for computer games, allowing customers to download and play games without restriction. Apple dropped DRM from iTunes music downloads in 2009 after determining that consumer frustration outweighed the benefits of digital restriction. These experiences suggest that trusting consumers can be a more effective business strategy than keeping them locked in closed digital ecosystems.[17]

Tor Books' experiment challenged the conventional wisdom that digital content needs to be locked down to be commercially viable. It demonstrated that DRM-free approaches can work within the publishing industry and highlighted important lessons for the ongoing debate regarding consumer rights, digital ownership, and content distribution. The experience showed that removing DRM doesn't necessarily increase piracy, can boost customer loyalty, and might actually be good business practice. While DRM remains the status quo in publishing, Tor's experience provides compelling evidence that more consumer-friendly approaches to digital content protection are both possible and potentially beneficial for all stakeholders.[18]

DRM and Libraries

Libraries face unique challenges regarding DRM, as their mission to provide free access to information often conflicts with the restrictions imposed by DRM systems. Libraries must balance respecting copyright holders' rights while fulfilling their core mission of providing access to information for all.[19]

When library patrons borrow e-books, DRM controls access to titles, limits loan periods, and restricts the number of simultaneous users. While DRM enables various business models, such as pay-per-use systems that allow libraries to pay only for accessed titles, it also creates significant

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Sander ed., 2002).

[17] Tobias Hauser & Christian Wenz, DRM Under Attack: Weaknesses in Existing Systems, in DIGITAL RIGHTS MANAGEMENT, LECTURE NOTES IN COMPUTER SCIENCE, vol. 2770, 206–23 (Eberhard Becker, Willms Buhse, Dirk Günnewig, & Niels Rump eds., 2003).

[18] Natali Helberger, Nicole Dufft, Margreet Groenenboom, Kristóf Kerényi, Carsten Orwat, & Ulrich Riehm, Digital Rights Management and Consumer Acceptability: A Multi-disciplinary Discussion of Consumer Concerns and Expectations: State of the Art Report—First Supplement, (INDICARE, May 2005).

[19] Denise M. Davis & Tim Lafferty, Digital Rights Management: Implications for Libraries, 15 BOTTOM LINE 23 (Mar. 2002).

limitations. For example, major publishers like Penguin Random House, HarperCollins, and Macmillan use DRM to enforce loan caps, typically allowing each e-book to be loaned only 52 times over two years before the library must renew its license.[20]

In academic libraries, DRM limits downloading, sharing, and printing options for students and researchers, hampering educational and research activities. The American Library Association and the Electronic Frontier Foundation have advocated against restrictive DRM for several reasons: it impedes fair use rights, limits user options, creates cumbersome user experiences, and prevents true ownership of digital materials.[21]

Library professionals have identified several key challenges posed by DRM. First, the control over DRM systems often resides with vendors rather than libraries, reducing local control and potentially increasing costs. Second, libraries may need to negotiate rights with each publisher, sometimes on a title-by-title basis, creating administrative burdens. Third, DRM raises concerns about long-term archiving and preservation of digital materials for future generations.[22]

The summer of 2019 saw a significant escalation in DRM-related conflicts when Macmillan announced an eight-week embargo on library e-books across all its imprints. Under this policy, libraries could purchase only one copy of a new e-book during the first eight weeks after publication, with additional copies becoming available only after this period. Macmillan claimed this decision was necessary to protect author royalties, stating, "What we were seeing was really reaching a tipping point, where we'd have to explain to our authors that while your readership is growing, your royalty statement will be getting smaller and smaller."[23] However, the publisher provided no concrete evidence to support claims of sales cannibalization.

The American Library Association immediately expressed concern about this decision, arguing that it contradicted the core mission of libraries to provide equitable access to information. This

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[20] American Library Association, Digital Rights Management, ALA/DCWG Tip Sheet, no. 1, 1 (July 2012)
[21] Jolanda-Pieta Van Arnhem & Lindsay Barnett, Is Digital Rights Management (DRM) Impacting E-book Adoption in Academic Libraries, 15 CHARLESTON ADVISOR 63, 63–65 (Jan. 2014).
[22] Karen Coyle, Rights Management and Digital Library Requirements, ARIADNE, no. 40 (July 30, 2004)
[23] John Sargent, News Release to Macmillan Authors, Macmillan Illustrators, and Agents, INFODOCKET (July 2019), https://www.infodocket.com/wp-content/uploads/2019/07/John-Sargent-Letter-Changes-to-Library-Lending-Terms.pdf.

controversy highlights the ongoing tension between publisher business models and the public service mission of libraries.[24]

Despite these challenges, there has been progress in both public and academic library settings. In academic libraries, publishers including Oxford University Press, Cambridge University Press, SAGE, Springer/Palgrave, Elsevier, Wiley, De Gruyter, Brill, and Emerald have begun providing DRM-free titles via their platforms. Even aggregators like EBSCO, ProQuest, JSTOR, and Project MUSE now offer DRM-free options. This shift has been driven partly by user feedback, with a 2018 Library Journal survey finding that 74% of students believe e-books should have no restrictions, 66% prefer using restriction-free e-books, and 37% use only restriction-free e-books when conducting research.[25]

The rise of the Open Access movement has also influenced DRM policies in academic publishing. Open Access initiatives have encouraged publishers to reconsider restrictive DRM practices, leading to more DRM-free content becoming available. A 2017 white paper by Springer Nature revealed that open access books enjoyed seven times more downloads, 50% more citations, and ten times more online mentions than paywalled titles, providing economic incentives for publishers to adopt more open approaches.[26]

Library information scientists have proposed several recommendations for navigating DRM challenges: embracing DRM technology with an open mind while recognizing its limitations, maintaining stringent privacy protections for patrons, educating users about available resources and access methods, and actively supporting open access initiatives through financial contributions and advocacy. By balancing these approaches, libraries can continue to fulfill their mission while respecting the legitimate interests of content creators and publishers.[27]

Effects of Piracy and DRM Effectiveness

[24] Benjamin J. Keele & Jere D. Odell, Managing Digital Rights in Open Access Works, in DIGITAL RIGHTS MANAGEMENT: THE LIBRARIAN'S GUIDE 103, 103–17 (Catherine A. Lemmer & Carla P. Wale eds., 2016).
[25] Mirela Roncevic, The Long and Winding Road to DRM-Free E-books in Academic Libraries, in NO SHELF REQUIRED 3: NEW ERA FOR E-BOOKS AND DIGITAL CONTENT (Mirela Roncevic & Peyton Stafford eds., forthcoming).
[26] Christina Emery, Mithu Lucraft, Agata Morka, & Ros Pine, The OA Effect: How Does Open Access Affect the Usage of Scholarly Books? 4 (Springer Nature, Nov. 2017, updated Jan. 2018).
[27] Edward Sanchez & Carrie Russell, Sci-Hub Unmasked: Piracy, Information Policy, and Your Library, 77 C. & RES. LIBR. NEWS 122, 122–25 (2016).

The relationship between DRM, piracy, and sales remains complex and often misunderstood. Research on the effects of piracy on sales has produced mixed results, challenging the assumption that strict DRM effectively prevents revenue loss.

According to a survey by Digimarc and Nielsen in 2017, e-book pirates come from diverse demographic groups—41% are aged 18-29, 47% are aged 30-44, and 13% are 45 or older. Contrary to popular belief, most pirates are not low-income consumers but members of the upper middle class. The survey also revealed that 44% of pirates also purchase e-books legally, suggesting that piracy is not always a substitute for legitimate purchases.[28]

In 2017, the UK government's Intellectual Property Office found that about 17% of e-books in the UK were consumed illegally—approximately four million books. While organizations like the Publishers Association view this as "4m books that authors and publishers aren't getting paid for," this perspective has been criticized for assuming that each pirated copy represents a lost sale.[29] This assumption lacks empirical support, as many who download pirated content would not have purchased it otherwise.

The effectiveness of DRM in combating piracy has been questioned by numerous studies. Haber and colleagues found in 2003 that DRM measures were not effective at preventing piracy because even if only a small fraction of users could transform content from protected to unprotected form, illegitimate distribution networks would make that content widely available.[30] They identified two types of piracy: unauthorized acquisition (obtaining content illegally) and unauthorized use (using legally obtained content in unauthorized ways). A fundamental flaw in DRM strategy has been assuming that addressing the second type would also solve the first.

Some studies have found positive effects of less restrictive approaches. Reimers's 2016 study published in the Journal of Law and Economics showed a 14% increase in e-book sales with certain anti-piracy measures, though it did not specifically examine DRM but rather techniques

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[28] Digimarc, Inside the Mind of a Book Pirate (study, Beaverton, OR: Digimarc, 2017).
[29] Intellectual Property Office, Online Copyright Infringement Tracker Survey: Latest Wave of Research (March 2017): Overview and Key Findings (Newport, UK: Intellectual Property Office, July 2017).
[30] Stuart Haber, Bill Horne, Joe Pato, Tomas Sander, & Robert Endre Tarjan, If Piracy Is the Problem, Is DRM the Answer?, in DIGITAL RIGHTS MANAGEMENT, LECTURE NOTES IN COMPUTER SCIENCE, vol. 2770, 224-25 (Eberhard Becker, Willms Buhse, Dirk Günnewig, & Niels Rump eds., 2003).

like delinking unauthorized copies based on metadata.[31] In the music industry, Laurina Zhang's 2013 dissertation found that removing DRM increased music sales by 10%, particularly for lower-selling albums, suggesting that relaxing DRM could potentially increase interest in backlist titles.[32]

The European Commission's reluctance to publish a €360,000 study on piracy (until forced by an information request) raises questions about political motivations in the piracy debate. The study, conducted by Dutch organization Ecorys, found that except for recently released blockbuster movies, there was no evidence that piracy affected sales of copyrighted materials or that online infringement displaced sales. For video games, the study actually showed that piracy helped boost sales.[33]

Authors have varied perspectives on DRM and piracy. Bestselling author Paulo Coelho deliberately leaked e-book versions of "The Alchemist" on piracy networks in Russia in 1999, and subsequently saw print sales increase from about 1,000 copies to over one million per year. He maintains: "The more people 'pirate' a book, the better. If the reader likes the excerpt of a book that he saw on the internet, he will most likely want to buy it the next day."[34] Coelho even created the website "Pirate Coelho" with download links to his books, claiming this increased rather than decreased sales.

In contrast, J.K. Rowling initially refused to make Harry Potter books available digitally due to piracy fears. Ironically, this didn't prevent piracy but may have encouraged it, as those wanting digital copies had no legitimate option. Later, Rowling's publisher began offering e-versions of Harry Potter books with watermarking DRM to track pirated copies to their source.[35]

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[31] Imke Reimers, Can Private Copyright Protection Be Effective? Evidence from Book Publishing, 59 J.L. & ECON. 412 (May 2016).

[32] Laurina Zhang, Intellectual Property Strategy and the Long Tail: Evidence from the Recorded Music Industry (PhD diss., Univ. of Toronto, 2013).

[33] Martin Van der Ende, Joost Poort, Robert Haffner, Patrick de Bas, Anastasia Yagafarova, Sophie Rohlfs, & Harry van Til, Estimating Displacement Rates of Copyrighted Content in the EU: Final Report (Brussels: European Commission, May 2015), https://cdn.netzpolitik.org/wp-upload/2017/09/displacement_study.pdf.

[34] Paolo Coelho, quoted in Anderson Antunes, 'Pirate My Books,' Urges the Wizard (Not Harry Potter!), FORBES MAG., July 29, 2011, https://www.forbes.com/sites/andersonantunes/2011/07/29/pirate-my-books-urges-the-wizard-not-harry-potter/#74460e526821.

[35] Martin Zimerman, E-books and Piracy: Implications/Issues for Academic Libraries, 112 NEW LIBR. WORLD 67, 67-75 (2011).

These varied perspectives and research findings suggest that the relationship between DRM, piracy, and sales is far more nuanced than often portrayed in policy discussions. Strict DRM may, in some cases, drive legitimate users to piracy due to frustration with restrictions, while more flexible approaches might actually increase sales by improving user experience and expanding market reach.

Policy Recommendations and Conclusion

Based on the extensive analysis of DRM's impact on consumer rights and the digital content ecosystem, this paper proposes a series of policy recommendations to restore balance between copyright protection and consumer rights while addressing the legitimate interests of all stakeholders.

Policymakers, legislators, and industry stakeholders should implement a more balanced approach to DRM that better aligns with consumer expectations and traditional rights of ownership. The current system has tilted too far in favor of copyright holders, creating unnecessary friction in the digital marketplace and potentially stifling innovation and cultural participation. The following recommendations offer a path forward:

First, policymakers should establish clear legal standards that protect basic consumer rights in digital environments. Legislation should require that DRM implementations preserve essential consumer rights, including the ability to make personal backup copies, transfer content between devices owned by the same user, and exercise fair use rights for educational, research, and critical purposes. These rights should be non-negotiable through EULAs or other contractual mechanisms, similar to how the European Union has established minimum consumer rights that cannot be overridden by contract.[36]

Second, transparency requirements should be significantly enhanced. Digital retailers should be required to clearly disclose all DRM restrictions before purchase, using standardized, easy-to-understand language. These disclosures should clearly distinguish between the purchase of a license and full ownership, ensuring consumers can make informed decisions. Current disclosure

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[36] L. Jean Camp, First Principles of Copyright for DRM Design, 7 INTERNET COMPUTING, IEEE 64, 64-66 (May–June 2003), https://doi.org/10.1109/MIC.2003.1200302.

practices are often buried in lengthy EULAs that few consumers read, creating a significant information asymmetry in the marketplace.[37]

Third, legal reforms should establish a digital first sale doctrine that allows consumers to resell or transfer their digital purchases under appropriate circumstances. The Capitol Records v. ReDigi decision highlighted the absence of such rights in the digital realm, but policy intervention could establish a balanced framework that respects both copyright protection and consumer expectations regarding ownership.[38] This reform would help align digital rights more closely with traditional ownership rights while establishing appropriate safeguards against abuse.

Fourth, policymakers should incentivize less restrictive DRM models through tax benefits, grants, or other market mechanisms. The experience of companies like Tor Books demonstrates that less restrictive approaches can be commercially viable while offering significant benefits to consumers and potentially expanding markets for creative content. Policy interventions could help accelerate industry adoption of more balanced approaches by offsetting perceived risks and highlighting potential benefits.[39]

Fifth, legal protection for libraries and educational institutions should be strengthened to ensure they can fulfill their public service missions in the digital age. Legislation should prevent publishers from imposing unreasonable restrictions on libraries, such as artificial scarcity through excessive embargoes or prohibitively expensive licensing terms. Libraries serve a crucial role in providing equitable access to information and preserving cultural heritage, and these functions should not be undermined by overly restrictive DRM policies.[40]

Finally, funding for research on DRM effects and alternative models should be increased to develop evidence-based policies. Current debates are often shaped more by assumptions than empirical evidence, as illustrated by the European Commission's reluctance to publish research findings that contradicted prevailing narratives about piracy. Independent research can help

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[37] Bruce Schneier, The Futility of Digital Copy Prevention, CRYPTO-GRAM (newsletter), May 15, 2001, https://www.schneier.com/crypto-gram/archives/2001/0515.html#3.
[38] Felix Oberholzer-Gee & Koleman Strumpf, The Effect of File Sharing on Record Sales: An Empirical Analysis, 115 J. POL. ECON. 1, 1–42 (2007), https://doi.org/10.1086/511995.
[39] Ed Felten, Why Unbreakable Codes Don't Make Unbreakable DRM, FREEDOM TO TINKER (blog), Dec. 3, 2002, https://freedom-to-tinker.com/2002/12/03/why-unbreakable-codes-dont-make-unbreakable-drm.
[40] Hasshi Sudler, Effectiveness of Anti-piracy Technology: Finding Appropriate Solutions for Evolving Online Piracy, 56 BUS. HORIZONS 149, 149–57 (2013).

identify approaches that balance stakeholder interests effectively while avoiding unintended consequences.[41]

[41] Stuart Haber, Bill Horne, Joe Pato, Tomas Sander & Robert Endre Tarjan, If Piracy Is the Problem, Is DRM the Answer?, in DIGITAL RIGHTS MANAGEMENT, LECTURE NOTES IN COMPUTER SCIENCE, vol. 2770, 224-31 (Eberhard Becker, Willms Buhse, Dirk Günnewig & Niels Rump eds., 2003).

Bibliography

Cases

Capitol Records, LLC v. ReDigi Inc., 934 F. Supp. 2d 640 (S.D.N.Y. 2013).

HarperCollins Publishers LLC v. Open Road Integrated Media, Inc., 754 F.3d 132 (2d Cir. 2014).

MDY Industries, LLC v. Blizzard Entertainment, Inc., 629 F.3d 928 (9th Cir. 2010).

Vernor v. Autodesk, Inc., 621 F.3d 1102 (9th Cir. 2010).

Articles, Books, and Reports

Amber Sami Kubesch & Stephen Wicker, Digital Rights Management: The Cost to Consumers, 103 PROC. OF THE IEEE 726 (2015).

American Library Association, Digital Rights Management, ALA/DCWG TIP SHEET, no. 1 (July 2012).

Anderson Antunes, 'Pirate My Books,' Urges the Wizard (Not Harry Potter!), FORBES MAG., July 29, 2011.

Benjamin J. Keele & Jere D. Odell, Managing Digital Rights in Open Access Works, in DIGITAL RIGHTS MANAGEMENT: THE LIBRARIAN'S GUIDE 103 (Catherine A. Lemmer & Carla P. Wale eds., 2016).

Bruce Schneier, The Futility of Digital Copy Prevention, CRYPTO-GRAM (newsletter), May 15, 2001.

Bruce Schneier, Quickest Patch Ever, WIRED, Sept. 7, 2006.

Christina Emery, Mithu Lucraft, Agata Morka & Ros Pine, THE OA EFFECT: HOW DOES OPEN ACCESS AFFECT THE USAGE OF SCHOLARLY BOOKS? (Springer Nature, Nov. 2017, updated Jan. 2018).

Chris Jay Hoofnagle, Digital Rights Management: Many Technical Controls on Digital Content Distribution Can Create a Surveillance Society, 5 COLUM. SCI. & TECH. L. REV. 1 (2004).

Cory Doctorow, Audio Version of Cory Doctorow's DRM Talk (talk to Microsoft Research Group at its Redmond offices on June 17, 2004).

Denise M. Davis & Tim Lafferty, Digital Rights Management: Implications for Libraries, 15 BOTTOM LINE 23 (Mar. 2002).

Digimarc, INSIDE THE MIND OF A BOOK PIRATE (Beaverton, OR: Digimarc, 2017).

Ed Felten, Why Unbreakable Codes Don't Make Unbreakable DRM, FREEDOM TO TINKER (blog), Dec. 3, 2002.

Edward Sanchez & Carrie Russell, Sci-Hub Unmasked: Piracy, Information Policy, and Your Library, 77 C. & RES. LIBR. NEWS 122 (2016).

Felix Oberholzer-Gee & Koleman Strumpf, The Effect of File Sharing on Record Sales: An Empirical Analysis, 115 J. POL. ECON. 1 (2007).

Frederick W. Dingledy & Alex Berrio Matamoros, What Is Digital Rights Management?, in DIGITAL RIGHTS MANAGEMENT: THE LIBRARIAN'S GUIDE 1 (Catherine A. Lemmer & Carla P. Wale eds., 2016).

Hasshi Sudler, Effectiveness of Anti-piracy Technology: Finding Appropriate Solutions for Evolving Online Piracy, 56 BUS. HORIZONS 149 (2013).

Imke Reimers, Can Private Copyright Protection Be Effective? Evidence from Book Publishing, 59 J.L. & ECON. 412 (May 2016).

Intellectual Property Office, ONLINE COPYRIGHT INFRINGEMENT TRACKER SURVEY: LATEST WAVE OF RESEARCH (MARCH 2017): OVERVIEW AND KEY FINDINGS (Newport, UK: Intellectual Property Office, July 2017).

Jolanda-Pieta Van Arnhem & Lindsay Barnett, Is Digital Rights Management (DRM) Impacting E-book Adoption in Academic Libraries, 15 CHARLESTON ADVISOR 63 (Jan. 2014).

Jonathan Zittrain, The Copyright Cage, LEGAL AFFAIRS, July/Aug. 2003.

Julie Crisp, One Year Later, the Results of Tor Books UK Going DRM-Free, TOR (Apr. 29, 2013).

Karen Coyle, Rights Management and Digital Library Requirements, ARIADNE, no. 40 (July 30, 2004).

Karen Coyle, The Technology or Rights: Digital Rights Management, Part 4, under "DRM and the Law" (based on a talk originally given at the Library of Congress, Nov. 19, 2003).

L. Jean Camp, DRM: Doesn't Really Mean Digital Copyright Management, KSG Working Paper Series RWP02-034 (Aug. 2002).

L. Jean Camp, First Principles of Copyright for DRM Design, 7 INTERNET COMPUTING, IEEE 64 (May–June 2003).

Laurina Zhang, Intellectual Property Strategy and the Long Tail: Evidence from the Recorded Music Industry (PhD diss., Univ. of Toronto, 2013).

Martin Van der Ende, Joost Poort, Robert Haffner, Patrick de Bas, Anastasia Yagafarova, Sophie Rohlfs & Harry van Til, ESTIMATING DISPLACEMENT RATES OF COPYRIGHTED CONTENT IN THE EU: FINAL REPORT (Brussels: European Commission, May 2015).

Martin Zimerman, E-books and Piracy: Implications/Issues for Academic Libraries, 112 NEW LIBR. WORLD 67 (2011).

Mir Mohammad Azad, Abu Hasnat Shohel Ahmed & Asadul Alam, Digital Rights Management, 10 INT'L J. COMPUT. SCI. & NETWORK SOC'Y 24 (2010).

Mirela Roncevic, The Long and Winding Road to DRM-Free E-books in Academic Libraries, in NO SHELF REQUIRED 3: NEW ERA FOR E-BOOKS AND DIGITAL CONTENT (Mirela Roncevic & Peyton Stafford eds., forthcoming).

Natali Helberger, Nicole Dufft, Margreet Groenenboom, Kristóf Kerényi, Carsten Orwat & Ulrich Riehm, DIGITAL RIGHTS MANAGEMENT AND CONSUMER ACCEPTABILITY: A MULTI-DISCIPLINARY DISCUSSION OF CONSUMER CONCERNS AND EXPECTATIONS: STATE OF THE ART REPORT—FIRST SUPPLEMENT (INDICARE, May 2005).

Priti Trivedi, Writing the Wrong: What the E-book Industry Can Learn from Digital Music's Mistakes with DRM, 18 J.L. & POL'Y 925 (2010).

Stefan Bechtold, From Copyright to Information Law: Implications of Digital Rights Management, in SECURITY AND PRIVACY IN DIGITAL RIGHTS MANAGEMENT, LECTURE NOTES IN COMPUTER SCIENCE, vol. 2320, 213 (Tomas Sander ed., 2002).

Stuart Haber, Bill Horne, Joe Pato, Tomas Sander & Robert Endre Tarjan, If Piracy Is the Problem, Is DRM the Answer?, in DIGITAL RIGHTS MANAGEMENT, LECTURE NOTES IN COMPUTER SCIENCE, vol. 2770, 224 (Eberhard Becker, Willms Buhse, Dirk Günnewig & Niels Rump eds., 2003).

Tim O'Reilly, Piracy Is Progressive Taxation, and Other Thoughts on the Evolution of Online Distribution, O'REILLY (2002).

Tobias Hauser & Christian Wenz, DRM Under Attack: Weaknesses in Existing Systems, in DIGITAL RIGHTS MANAGEMENT, LECTURE NOTES IN COMPUTER SCIENCE, vol. 2770, 206 (Eberhard Becker, Willms Buhse, Dirk Günnewig & Niels Rump eds., 2003).

Truls Fretland, Lothar Fritsch & Arne-Kristian Groven, STATE OF THE ART IN DIGITAL RIGHTS MANAGEMENT, MARIAGE Project Report 1018 (Norsk Regnesentral [Norwegian Computing Center], 2008).

Online Sources

John Sargent, News Release to Macmillan Authors, Macmillan Illustrators, and Agents, INFODOCKET (July 2019), https://www.infodocket.com/wp-content/uploads/2019/07/John-Sargent-Letter-Changes-to-Library-Lending-Terms.pdf.